# CS-303 Assignment 3

(25 points)

Q1. (13 points) Implement the Queue data structure (you can use linked list or circular array for implementation) with offer(), poll(), peek(), size() and empty() functions. The queue should be a template class (should work with any data type). Perform the following operations:

a) Instantiate the queue with integers and push 10 values into the queue
b) Display all the elements in a queue using just the queue functions mentioned above (offer, poll, peek, size etc.)
c) Write a new queue function called move_to_rear that moves the element currently at the front of the queue to the rear of the queue. The element that was second in line will be the new front element. Do this using functions offer, peek, and poll.

Q2. (5 points) Write a recursive **linear search function** with a recursive step that finds the last occurrence of a target in a ArrayList, not the first.

Q3. (7 points) Modify the source code for the insertion sort so that it can sort a list (linked list or you can also use queue) of integers (You can find the source code on lecture 10 slide 147).

**Submission guidelines:**

1. All the functionality of the program should be implemented as functions and methods.
2. The code should be well commented
3. Create a report (readme file) that contains instruction on how to run the code and screen shots of the outputs
4. Upload your report and code files to GitHub.
5. Submit the GitHub link on Canvas by due date.